

Full configuration interaction: variational principle, Hamiltonian matrix, diagrams, iteration – and a first step into Hartree-Fock theory

FYS4480/FYS9480 – Week 38, September 17–18, 2026

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Thursday September 17: Slater determinants as bit patterns; the pairing model as our first FCI model; the FCI equation from variational calculus (Ritz), the eigenvalue problem and the characteristic polynomial; the structure of the Hamiltonian matrix and the Condon-Slater rules; truncations and size consistency

Friday September 18: the projected FCI equations, their diagrams, the “non-practical” iterative solution and its link to Hartree-Fock; Hartree-Fock theory (chapter 6): mean field, variational calculus, the Hartree-Fock equations, the self-consistent field

Reading: these slides; lecture notes chapter 3 (section 3.16), chapter 5 (all, in particular 5.3–5.4 and 5.8–5.10) and chapter 6 (sections 6.1–6.5); Szabo and Ostlund chapters 3–4; Shavitt and Bartlett chapters 4 and 9. Code: `WeeklySlides/week38.ipynb`, `BookPrograms/chapter05/fci.py`, `chapter06/hartreefock.py`

Plan for week 38

Thursday September 17, 2×45 min

- Where week 37 stopped (5 min): the normal-ordered Hamiltonian and the first FCI ideas were covered; bit patterns, truncations and size consistency were not
- Slater determinants as bit patterns, $\hat{H}|\Phi\rangle$ by bit operations – the inner loop of every FCI code (section 3.16); the pairing model as our first FCI model, 70 determinants against 6 (section 5.6)
- The FCI equation properly derived: variational calculus with a Lagrange multiplier, Ritz's method as an optimisation of the coefficients, the eigenvalue problem, $\det(H - E1) = 0$ and its characteristic polynomial (section 5.3)
- The structure of the Hamiltonian matrix in particle-hole language and the Condon-Slater rules of weeks 36–37 (section 5.4); truncations and size consistency (sections 5.8–5.9)

Friday September 18, 2×45 min

- The FCI equation projected: energy, singles and doubles equations, and their diagrammatic representation
- The non-practical way of solving the FCI equations: the amplitude equations by iteration, and what that teaches about perturbation theory, coupled cluster and Hartree-Fock (section 5.10, `fci.do.txt`)
- Hartree-Fock theory (chapter 6): mean field, variational calculus, determinants under a change of basis, the Hartree-Fock equations, the density matrix and the self-consistent field, Brillouin's theorem (sections 6.1–6.5); exercises (last frames)

Thursday, lecture 1: Bit patterns; the pairing model by FCI

Where week 37 ended

Thursday of week 37 gave us the Hamiltonian in the form we use for the rest of the course,

$$\hat{H} = E_{\text{ref}} + \sum_{pq} f_{pq} \{a_p^\dagger a_q\} + \frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle \{a_p^\dagger a_q^\dagger a_s a_r\}, \quad f_{pq} = \langle p|\hat{h}_0|q\rangle + \sum_i \langle pi|\hat{v}|qi\rangle,$$

and the three numbers that follow from it by the generalised Wick theorem: $\langle c|\hat{H}|c\rangle = E_{\text{ref}}$, $\langle \Phi_i^a|\hat{H}|c\rangle = f_{ai}$, $\langle \Phi_{ij}^{ab}|\hat{H}|c\rangle = \langle ab|\hat{v}|ij\rangle$, and zero beyond. Friday introduced full configuration interaction and started on the pairing model.

What was *not* reached, and what we do this week:

- the **bit representation** of Slater determinants (section 3.16) – how the formalism becomes code (today, first half);
- the pairing model as a complete first FCI calculation, and the **eigenvalue problem derived properly**: variational calculus, the Ritz method, the characteristic polynomial (today, second half);
- the **structure of the Hamiltonian matrix** tied to the Condon-Slater rules, **truncations and size consistency** (today, end);
- tomorrow: the projected FCI equations as **diagrams**, their iterative solution, and the beginning of **Hartree-Fock theory**.

Reminder: the Condon-Slater rules in the language of week 37

For two determinants brought to maximum coincidence ($\hat{H} = \hat{H}_0 + \hat{V}$, antisymmetrised elements):

determinants differ in	$\langle \Phi \hat{H}_0 \Phi' \rangle$	$\langle \Phi \hat{V} \Phi' \rangle$	with $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$, relative to $ \Phi_0\rangle$
no orbital	$\sum_i h_{ii}$	$\frac{1}{2} \sum_{ij} \langle ij v ij \rangle$	E_{ref}
one orbital, $i \rightarrow a$	h_{ai}	$\sum_j \langle aj v ij \rangle$	f_{ai}
two orbitals, $ij \rightarrow ab$	0	$\langle ab v ij \rangle$	$\langle ab v ij \rangle$
three or more	0	0	0

- Week 36 derived the first three columns with the generalised Wick theorem (four surviving pairings); week 37 read the last column off the normal-ordered Hamiltonian.
- Two determinants that are *both* excitations of $|\Phi_0\rangle$ obey the same rules, applied to the orbitals in which they differ from each other: $\langle \Phi_i^a | \hat{H} | \Phi_j^b \rangle = \delta_{ij} f_{ab} - \delta_{ab} f_{ji} + \langle aj | v | ib \rangle$ (week 37, Example 4) is the one-orbital rule dressed by the mean field.

Many-body connection

Everything in this week's FCI matrix is one of these four entries. The bit representation is nothing but a fast way of finding out *which* of the four, and with what sign.

From the formalism to a running code (section 3.16)

The time-consuming part of any shell-model or FCI code is the action of the Hamiltonian on a determinant,

$$\left(\sum_{pq} \langle p | \hat{h}_0 | q \rangle a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle a_p^\dagger a_q^\dagger a_s a_r \right) a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \cdots a_{\alpha_N}^\dagger |0\rangle.$$

A practically useful way to implement it: **encode a Slater determinant as a bit pattern**. With n single-particle states $\alpha_0, \dots, \alpha_{n-1}$ and $N \leq n$ particles, an occupied state is a 1, an empty one a 0. For $n = 16$ and particles in $\alpha_3, \alpha_6, \alpha_{10}, \alpha_{13}$:

$$\Phi_{3,6,10,13} = a_{\alpha_3}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = |0001001000100100\rangle, \quad 2^3 + 2^6 + 2^{10} + 2^{13} = 9288.$$

- The ordering of the states is important: it defines the order of the creation operators, and hence the sign convention – the dictionary of section 3.1 with a fixed ordering.
- There are 2^n bit patterns but only $\binom{n}{N}$ determinants with N particles: $\dim \mathcal{H} = \binom{n}{N}$ – 1820 for $n = 16$, $N = 4$.
- The Fock-space matrices of the notebooks *are* this representation; today we never form the $2^n \times 2^n$ matrices, only act on one pattern at a time.

The action of a creation operator on a bit pattern

$a_{\alpha_4}^\dagger \Phi_{3,6,10,13} = a_{\alpha_4}^\dagger a_{\alpha_3}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = -a_{\alpha_3}^\dagger a_{\alpha_4}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = -|0001101000100100\rangle$,
 one anticommutation to put $a_{\alpha_4}^\dagger$ in its place, while $a_{\alpha_6}^\dagger \Phi_{3,6,10,13} = -a_{\alpha_3}^\dagger (a_{\alpha_6}^\dagger)^2 a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = 0$.

The recipe

- If bit b_j is already 1 and we act with $a_{\alpha_j}^\dagger$: return the null vector (Pauli).
- If $b_j = 0$: set it to 1 and return the sign factor $(-1)^l$, l the number of bits set *before* bit j (the creation operators that $a_{\alpha_j}^\dagger$ has to pass).

For a_{α_j} : return zero if $b_j = 0$; otherwise clear the bit and return $(-1)^l$ with the same l .

$a_{\alpha_2}^\dagger 00111\rangle$	$= 0$	$a_{\alpha_2}^\dagger 10101\rangle$	$= 0$
$a_{\alpha_2}^\dagger 01011\rangle$	$= (-1) 01111\rangle$	$a_{\alpha_2}^\dagger 10110\rangle$	$= 0$
$a_{\alpha_2}^\dagger 01101\rangle$	$= 0$	$a_{\alpha_2}^\dagger 11001\rangle$	$= (+1) 11101\rangle$
$a_{\alpha_2}^\dagger 10011\rangle$	$= (-1) 10111\rangle$	$a_{\alpha_2}^\dagger 11010\rangle$	$= (+1) 11110\rangle$

(bits written $\alpha_0\alpha_1\alpha_2\alpha_3\alpha_4$ from left to right, as in the book; section 1 of the notebook reproduces the table).

The phase without a loop: bit masks

Consider $a_{\alpha_4}^\dagger a_{\alpha_0} \Phi_{0,3,6,10,13}$. Removing α_0 is a logical AND with the complement of the word $S_1 = |1000 \dots\rangle$; adding α_4 is a logical OR with $S_2 = |00001000 \dots\rangle$, after checking that the AND of S_2 with the state is zero. The phase of the pair move $a_{\alpha_j}^\dagger a_{\alpha_i}$ is

$$(-1)^{\#\{\text{occupied states strictly between } \alpha_i \text{ and } \alpha_j\}},$$

because the signs from removing α_i and from inserting α_j share all the occupied states outside the interval, which cancel in pairs. The mask of the states between α_i and α_j is integer arithmetic, the AND with the state selects the occupied ones, and a **bit-count** (popcount) gives l – here $|0001000 \dots\rangle$, $l = 1$, phase -1 .

```
def move(state, i, j): # a_j^dagger a_i acting on a bit pattern
    if not (state >> i) & 1: return 0, None # i empty: null vector
    if i == j: return 1, state # a_i^dagger a_i counts the occupation
    if (state >> j) & 1: return 0, None # j occupied: null vector
    lo, hi = min(i, j), max(i, j)
    between = state & ((1 << hi) - (1 << (lo + 1))) # occupied states strictly between
    phase = -1 if bin(between).count("1") % 2 else 1
    return phase, (state & ~(1 << i)) | (1 << j)
```

Section 1 of the notebook checks move against two successive single-operator moves for all 32×25 cases of a five-bit word.

$\hat{H}|\Phi\rangle$ by bit operations: the inner loop of every FCI code

Apply the operator string of each term of $\hat{H}$ to the pattern, right to left, and collect {new pattern: amplitude}. Only *occupied* q, r, s contribute, so those loops run over the N occupied orbitals, not over all n :

```
def apply_hamiltonian(state, h, vAS):
    out = {}
    occ = [p for p in range(n) if (state >> p) & 1]
    for q in occ: # sum_pq h_pq a_p^+ a_q
        for p in range(n):
            sign, new = apply_string(state, [(p, True), (q, False)])
            if sign: out[new] = out.get(new, 0.0) + sign * h[p, q]
    for r in occ: # 1/4 sum <pq|v|rs> a_p^+ a_q^+ a_s a_r
        for s in occ:
            if s == r: continue
            for p in range(n):
                for q in range(n):
                    if vAS[p, q, r, s] != 0.0:
                        sign, new = apply_string(state, [(p, True), (q, True), (s, False), (r, False)])
                        if sign: out[new] = out.get(new, 0.0) + 0.25 * sign * vAS[p, q, r, s]
    return out
```

Cost per determinant $\mathcal{O}(Nn + N^2n^2)$ bit operations. Section 1 of the notebook compares with the Fock-space matrices of week 37 for every pattern of a small space: identical, sign for sign. The FCI code of section 2 is a loop over this function.

Pause to think: bit patterns by hand

Pause to think #1

(about 4 min – discuss with your neighbour)

Five single-particle states $\alpha_0, \dots, \alpha_4$, bits written $\alpha_0\alpha_1\alpha_2\alpha_3\alpha_4$.

1. Evaluate $a_{\alpha_3}^\dagger a_{\alpha_0} |10110\rangle$ and $a_{\alpha_1}^\dagger a_{\alpha_4} |10011\rangle$, with signs, first by the definition (anticommute the operators into place) and then by the “occupied states strictly between” rule.
2. A two-body term $a_{\alpha_p}^\dagger a_{\alpha_q}^\dagger a_{\alpha_s} a_{\alpha_r}$ acts on a pattern with N bits set. For how many index quadruples (p, q, r, s) is the result non-zero, and how does this compare with the n^4 terms of the double sum? Take $n = 40$, $N = 8$.

Answer to pause #1

1. $|10110\rangle = a_0^\dagger a_2^\dagger a_3^\dagger |0\rangle$. a_0 removes the first operator with no sign, leaving $a_2^\dagger a_3^\dagger |0\rangle$; $a_3^\dagger$ then finds α_3 occupied: **zero**. (The rule agrees: $j = 3$ is occupied, null vector.)
 $|10011\rangle = a_0^\dagger a_3^\dagger a_4^\dagger |0\rangle$: a_4 must pass two operators, $(-1)^2$, leaving $a_0^\dagger a_3^\dagger |0\rangle$; $a_1^\dagger$ passes one, $(-1)^1$: result $-|11010\rangle$. By the rule: between α_1 and α_4 the occupied state is α_3 only, $l = 1$, phase -1 . ✓
2. r and s must be occupied and distinct: $N(N-1)$ ordered pairs; p and q must be empty after the removal or equal to r, s – at most $(n-N+2)(n-N+1)$ ordered pairs. For $n = 40$, $N = 8$: $56 \times 34 \cdot 33 \approx 6.3 \times 10^4$ against $40^4 = 2.56 \times 10^6$, a factor of 40. This is why the loops over r, s run over the occupied orbitals only, and why sparse storage of $\langle pq|v|rs\rangle$ (only the non-zero elements, grouped by symmetry) is the next thing a production code does.

Full CI in two lines (sections 5.1–5.2, week 37)

Reference $|\Phi_0\rangle = \prod_{i \leq F} a_i^\dagger |0\rangle$; every other determinant is an excitation of it, $|\Phi_i^a\rangle = a_a^\dagger a_i |\Phi_0\rangle$, $|\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle$, $\dots$, and the exact state is a superposition of all of them:

$$|\Psi_0\rangle = \sum_{PH} C_H^P |\Phi_H^P\rangle = \left(1 + \sum_{ai} C_i^a a_a^\dagger a_i + \sum_{\substack{a < b \\ i < j}} C_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i + \dots\right) |\Phi_0\rangle = (1 + \hat{C}) |\Phi_0\rangle,$$

with intermediate normalisation $C_0 = 1$ when convenient, or $\sum_{PH} |C_H^P|^2 = 1$ when we diagonalise. H stands for a set of hole indices, P for a set of particle indices.

The idea

If the single-particle basis describes the system well enough, diagonalising $\hat{H}$ in the space of *all* its determinants is exact. Full CI is the benchmark for every approximation of the course – which is why we do it first, and on a model where we can.

- In code a determinant is an integer; the expansion is a vector of length $\binom{n}{N}$ indexed by the sorted list of patterns; $\hat{H}$ acting on the vector is `apply_hamiltonian` applied to every pattern.
- The rest of today: *why* diagonalisation is the answer (the variational principle), what the matrix looks like, and what happens when we cannot afford all of it.

The pairing model as our first FCI model (section 5.6)

Four doubly degenerate levels $p = 1, \dots, 4$, energies $\varepsilon_p = (p - 1)\xi$, spin $\sigma = \pm$, four particles, $\xi = 1$:

$$\hat{H} = \xi \sum_{p\sigma} (p - 1) a_{p\sigma}^\dagger a_{p\sigma} - \frac{g}{2} \sum_{pq} a_{p+}^\dagger a_{p-}^\dagger a_{q-} a_{q+}, \quad \langle p + p - | \hat{v} | q + q - \rangle = -\frac{g}{2}, \text{ else } 0.$$

Reference $|\Phi_0\rangle$: levels 1 and 2 filled. The full space has $\binom{8}{4} = 70$ determinants, 1, 16, 36, 16, 1 over the excitation levels, and the block table computed by the code (section 2 of the notebook) is

	0p0h	1p1h	2p2h	3p3h	4p4h
0p0h	×	0	×	0	0
1p1h	0	×	0	×	0
2p2h	×	0	×	0	×
3p3h	0	×	0	×	0
4p4h	0	0	×	0	×

142 non-zero elements out of 4900.

- Here even the odd blocks are empty: the interaction moves a pair and never breaks one, so the **seniority** (number of unpaired particles) is conserved – a six-dimensional $s = 0$ block, an $s = 2$ block, an $s = 4$ block. The ideal first test: small enough for the full space, and the seniority-zero answer of chapter 4 is known.

The seniority-zero matrix, by the rules of week 37

Label a seniority-zero configuration by the pair of occupied levels: $|12\rangle, |13\rangle, |14\rangle, |23\rangle, |24\rangle, |34\rangle$. The diagonal element of $|pq\rangle$ is $2\xi[(p-1) + (q-1)] - g$ (two particles per level; the $p = q$ terms of the interaction give $-g/2$ for each of the two pairs). One application of the interaction moves *one* pair, so two configurations are connected by $-g/2$ if and only if they share exactly one level:

$$H = \begin{pmatrix} 2-g & -g/2 & -g/2 & -g/2 & -g/2 & 0 \\ -g/2 & 4-g & -g/2 & -g/2 & 0 & -g/2 \\ -g/2 & -g/2 & 6-g & 0 & -g/2 & -g/2 \\ -g/2 & -g/2 & 0 & 6-g & -g/2 & -g/2 \\ -g/2 & 0 & -g/2 & -g/2 & 8-g & -g/2 \\ 0 & -g/2 & -g/2 & -g/2 & -g/2 & 10-g \end{pmatrix}$$

- Every entry is a Condon-Slater element: $\langle \Phi_0 | \hat{H} | \Phi_0 \rangle = E_{\text{ref}} = 2 - g$;
 $\langle \Phi_{ij}^{ab} | \hat{H} | \Phi_0 \rangle = \langle ab | \hat{v} | ij \rangle = -g/2$ for a pair moved as a whole; $|12\rangle$ and $|34\rangle$ differ by *four* orbitals: zero. The diagonal $2p2h$ elements are Example 4 of week 37, e.g. $6 - g$ for $|23\rangle$.
- The three unconnected pairs, $|12\rangle$ – $|34\rangle$, $|13\rangle$ – $|24\rangle$, $|14\rangle$ – $|23\rangle$, are the zeros in the matrix.

Results: 70×70 against 6×6

g	E_{ref}	E_0 (6×6)	E_0 (70×70)	broken-pair amplitude
-1.00	3.000000	2.77987014	2.77987014	$< 10^{-15}$
-0.50	2.500000	2.43688426	2.43688426	$< 10^{-15}$
0.00	2.000000	2.00000000	2.00000000	0
0.50	1.500000	1.41677428	1.41677428	$< 10^{-15}$
1.00	1.000000	0.63554847	0.63554847	$< 10^{-15}$

- Identical to every digit. The 64 determinants with a broken pair are present in the basis but carry no amplitude in the ground state: the symmetry that chapter 4 imposed by hand is **recovered automatically** by the diagonalisation.
- This is the general situation. A symmetry of $\hat{H}$ block-diagonalises the FCI matrix, and one may exploit it (a smaller matrix, a labelled spectrum) or ignore it (a simpler code, a larger matrix). Production codes exploit it; `fci.py` does not, so that the two answers can be compared.
- The correlation energy $E_0 - E_{\text{ref}}$ is -0.083 at $g = 0.5$ and -0.364 at $g = 1$: 36% of the reference energy. At small $|g|$ it is quadratic, and the coefficient can be computed by hand (pause #2) – we shall meet it again tomorrow as the first iterate of the amplitude equations.

Pause to think: the correlation energy at small g

Pause to think #2

(about 4 min – discuss with your neighbour)

With the reference $|12\rangle$, second-order perturbation theory gives

$$E^{(2)} = \sum_{\Phi \neq \Phi_0} \frac{|\langle \Phi_0 | \hat{H}_I | \Phi \rangle|^2}{E_0^{(0)} - E_\Phi^{(0)}}.$$

1. Which configurations contribute, with what matrix element and which unperturbed excitation energies? Evaluate $E^{(2)}$.
2. Compare with the exact correlation energies -0.01954 ($g = 0.25$), -0.0832 ($g = 0.5$) and -0.3645 ($g = 1$). What do you conclude about the higher orders?

Answer to pause #2

1. The four configurations reachable by moving one pair, $|13\rangle, |14\rangle, |23\rangle, |24\rangle$, each with matrix element $-g/2$ and unperturbed excitation energies 2, 4, 4, 6:

$$E^{(2)} = \frac{g^2}{4} \left(\frac{1}{-2} + \frac{1}{-4} + \frac{1}{-4} + \frac{1}{-6} \right) = -\frac{7}{24} g^2 = -0.2917 g^2.$$

2. -0.01823 against -0.01954 at $g = 0.25$, -0.0729 against -0.0832 at $g = 0.5$, -0.2917 against -0.3645 at $g = 1$: the leading order is right at small coupling and misses 20% at $g = 1$. The higher orders are negative for $g > 0$ and their sum is what FCI delivers; perturbation theory computes them one order at a time, FCI all at once – and tomorrow's iteration of the FCI equations produces exactly this number as its first step.

Summary of lecture 1, and what comes next

- A Slater determinant is an integer; $a^\dagger$ and a are bit operations with a popcount for the sign; $\hat{H}|\Phi\rangle$ is a loop over the occupied orbitals and the non-zero matrix elements. Everything we compute this week runs on this.
- The pairing model by FCI: 70 determinants, 142 non-zero elements, one ground-state energy – identical to the 6×6 seniority-zero calculation to every digit, the symmetry being recovered by the diagonalisation.
- The correlation energy is $-\frac{7}{24}g^2$ at leading order and -0.3645 at $g = 1$: the gap between the two is what the higher orders, and the rest of the course, are about.

Many-body connection

We have *used* diagonalisation without justifying it. The next 45 minutes derive the FCI equation from the variational principle, show that the Ritz method is an optimisation over the expansion coefficients, and turn the equation into a matrix, a determinant and a characteristic polynomial – before looking at what the matrix is made of.

Thursday, lecture 2: The FCI equation; the Hamiltonian matrix

The variational principle, and the Ritz method (section 5.3)

For any normalisable trial state, $E[\Psi] = \langle \Psi | \hat{H} | \Psi \rangle / \langle \Psi | \Psi \rangle \geq E_0$ (week 34). Ritz's idea: choose the trial state as a **linear combination of fixed basis states** and optimise the *coefficients* only. With the orthonormal determinants $|\Phi_K\rangle$, $K = (P, H)$,

$$|\Psi\rangle = \sum_K C_K |\Phi_K\rangle, \quad E[C] = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \frac{\sum_{KL} C_K^* H_{KL} C_L}{\sum_K |C_K|^2}, \quad H_{KL} = \langle \Phi_K | \hat{H} | \Phi_L \rangle,$$

the **Rayleigh quotient**. The energy is now an ordinary function of $\dim \mathcal{H}$ complex numbers, and the problem is one of optimisation:

Ritz's method

Find the coefficients C that make $E[C]$ stationary. The stationary values are upper bounds to the exact eigenvalues, and the bounds can only improve when basis states are added.

- Two ways to do it: minimise $E[C]$ directly (a gradient method on the Rayleigh quotient – section 3 of the notebook does this), or minimise the numerator at fixed denominator with a **Lagrange multiplier**. The second is the classic route and gives the equation in closed form.
- Full CI is the Ritz method in the space of *all* determinants of the single-particle basis. Every truncation later today is the Ritz method in a subspace.

The derivation: variation with a Lagrange multiplier

Minimise $\langle \Psi_0 | \hat{H} | \Psi_0 \rangle$ subject to $\langle \Psi_0 | \Psi_0 \rangle = 1$, i.e. make

$$\mathcal{L}[C, C^*] = \langle \Psi_0 | \hat{H} | \Psi_0 \rangle - \lambda (\langle \Psi_0 | \Psi_0 \rangle - 1) = \sum_{PH} \sum_{P'H'} C_H^{P*} \langle \Phi_H^P | \hat{H} | \Phi_{H'}^{P'} \rangle C_{H'}^{P'} - \lambda \left(\sum_{PH} C_H^{P*} C_H^P - 1 \right)$$

stationary. Vary the coefficients, $C \rightarrow C + \delta C$, and keep the first-order terms:

$$\delta \mathcal{L} = \sum_{PH} \sum_{P'H'} \left\{ \delta[C_H^{P*}] \langle \Phi_H^P | \hat{H} | \Phi_{H'}^{P'} \rangle C_{H'}^{P'} + C_H^{P*} \langle \Phi_H^P | \hat{H} | \Phi_{H'}^{P'} \rangle \delta[C_{H'}^{P'}] - \lambda (\delta[C_H^{P*}] C_{H'}^{P'} + C_H^{P*} \delta[C_{H'}^{P'}]) \delta_{PP'} \delta_{HH'} \right\} = 0.$$

- The real and imaginary parts of C are two independent real variations; it is equivalent, and more convenient, to treat C and C^* as independent. Since $\delta[C^*]$ and $\delta[C]$ are complex conjugates of each other and $\hat{H}$ is Hermitian, the two groups of terms are complex conjugates, and it is necessary and sufficient that the coefficient of every $\delta[C_H^{P*}]$ vanishes:

$$\sum_{P'H'} \langle \Phi_H^P | \hat{H} | \Phi_{H'}^{P'} \rangle C_{H'}^{P'} = \lambda C_H^P \quad \text{for every set } PH.$$

- Multiply by C_H^{P*} and sum over PH : $\sum C^* H C = \lambda \sum |C|^2 = \lambda$, so **the multiplier is the energy**, $\lambda = E$.

The full CI equation

The full CI equation (5.3)

$$\sum_{P' H'} \langle \Phi_H^P | \hat{H} - E | \Phi_{H'}^{P'} \rangle C_{H'}^{P'} = 0 \quad \text{for every } PH \quad \Longleftrightarrow \quad HC = EC.$$

- The same result in one line: $(\hat{H} - E)|\Psi_0\rangle = 0$ projected onto every $\langle \Phi_H^P |$. The variational derivation tells us *more*: the projected equations are the stationarity conditions of the energy, so diagonalisation is the **variational answer**, not merely a convenience.
- A Hermitian eigenvalue problem of dimension $\binom{n}{N}$: the lowest eigenvalue is the best upper bound to E_0 in the space and can only go down when the space grows (next slide); the eigenvector gives every C_H^P at once.
- Matrix elements from the Condon-Slater rules; Householder-QL for small matrices, Lanczos for large ones – Lanczos needs only $\hat{H}|\Phi\rangle$, this morning's bit code, and its lowest Ritz value is the same bound computed in a Krylov space.
- The pattern – a constrained variation produces an eigenvalue problem, and the multiplier enforcing normalisation is the eigenvalue – returns tomorrow in Hartree-Fock theory, with the orbitals instead of the coefficients as the unknowns.

The optimisation view: gradient, Hessian, bounds

Treat $E[C]$ as a function to be minimised. For real coefficients, with $E = E[C]$,

$$\frac{\partial E}{\partial C_K} = \frac{2 [HC - E C]_K}{C^T C}, \quad \left. \frac{\partial^2 E}{\partial C_K \partial C_L} \right|_{C=C_0} = \frac{2 [H - E_0 1]_{KL}}{C_0^T C_0} \quad \text{at an eigenvector } C_0.$$

- The gradient vanishes *exactly* at the eigenvectors, and nowhere else: stationary points of the Rayleigh quotient = eigenpairs of H (section 3 of the notebook: $|\nabla E| < 10^{-14}$ at the six eigenvectors of the pairing matrix, never below 0.36 for 2000 random vectors).
- The Hessian at the lowest eigenvector is $H - E_0 1 \geq 0$: the ground state is a **minimum**, every other eigenvector a saddle point. Steepest descent on $E[C]$ converges to $E_0 = 0.63554847$ (the notebook, 500 steps) – slowly; Lanczos is the smart version of the same descent.
- **Interlacing (Hylleraas-Undheim-MacDonald)**. Order the Ritz values $E_1^{(m)} \leq \dots \leq E_m^{(m)}$ in an m -dimensional space. Then $E_k^{(m+1)} \leq E_k^{(m)}$ for every k and $E_k^{(m)} \geq E_k^{\text{exact}}$: every Ritz value is an upper bound to the corresponding exact level and decreases monotonically as the space grows. This is why CIS, CID, CISD, ..., FCI form a monotone sequence, for the ground state *and* for excited states.

From the equation to the matrix, the determinant and the polynomial

Order the determinants, $|\Phi_K\rangle$, $K = 1, \dots, d$, $d = \binom{n}{N}$; the FCI equation is d linear homogeneous equations for the d unknowns C_K :

$$\sum_{L=1}^d (H_{KL} - E \delta_{KL}) C_L = 0, \quad K = 1, \dots, d.$$

A homogeneous linear system has a non-trivial solution if and only if its determinant vanishes:

The secular equation

$$\det(\mathbf{H} - E \mathbf{1}) = \begin{vmatrix} H_{11} - E & H_{12} & \cdots & H_{1d} \\ H_{21} & H_{22} - E & \cdots & H_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ H_{d1} & H_{d2} & \cdots & H_{dd} - E \end{vmatrix} = (-1)^d p_d(E) = 0.$$

Here $p_d(E) = E^d - (\text{tr } \mathbf{H})E^{d-1} + \cdots + (-1)^d \det \mathbf{H}$ is the **characteristic polynomial** of $\mathbf{H}$.

- $p_d(E)$ has degree d ; its d roots are the eigenvalues, all real since $\mathbf{H}$ is Hermitian, and for each root the linear system yields the eigenvector. This is the algebraic content of “diagonalise”. The coefficient of E^{d-1} is minus the sum of the diagonal Condon-Slater elements, the constant term is $\det \mathbf{H}$, the rest are sums of principal minors; for $d \geq 5$ there is no formula for the roots.

The polynomial for the pairing model, and why we never use it

Two levels, one pair ($|1\rangle, |2\rangle$, diagonal $-g/2$ and $2\xi - g/2$, off-diagonal $-g/2$):

$$\begin{vmatrix} -\frac{g}{2} - E & -\frac{g}{2} \\ -\frac{g}{2} & 2\xi - \frac{g}{2} - E \end{vmatrix} = \left(E - \xi + \frac{g}{2}\right)^2 - \xi^2 - \frac{g^2}{4} = 0 \implies E_{\pm} = \xi - \frac{g}{2} \pm \sqrt{\xi^2 + \frac{g^2}{4}},$$

$E_- = -0.61803399$ at $\xi = g = 1$ – the golden ratio in disguise, and the number behind the size-consistency table in Thursday's lecture.

Four levels, seniority zero, $g = 1$ (section 3 of the notebook, `numpy.poly`):

$$p_6(E) = E^6 - 30E^5 + 352E^4 - 2038E^3 + 5979E^2 - 7940E + 3100,$$

with roots 0.635548, 2.935381, 5, 5, 7.208940, 9.220130, agreeing with `eigh` to 3×10^{-13} ;
 $\text{tr } H = 36 - 6g = 30$ and $\det H = 3100$ as they must.

- **Seventy determinants.** The coefficients of p_{70} span 55 orders of magnitude (9.6×10^{54} at the largest); the roots computed from the *exact* coefficients are off by 13 (!), and a relative perturbation of 10^{-14} of the coefficients moves them by as much – while the same perturbation of the *matrix* moves the eigenvalues by 4×10^{-13} .
- The eigenvalues are perfectly conditioned (Hermitian matrix: $|\delta E| \leq \|\delta H\|$), the polynomial roots are not. The characteristic polynomial is the **definition** of the eigenvalue problem, never its **algorithm**: Householder-QL and Lanczos work on the matrix directly and never form p_d .

Pause to think: the 2×2 problem by hand

Pause to think #3

(about 4 min – discuss with your neighbour)

One pair in two levels, $H = \begin{pmatrix} -g/2 & -g/2 \\ -g/2 & 2\xi - g/2 \end{pmatrix}$ in the basis $|1\rangle, |2\rangle$.

1. Write down the Rayleigh quotient for the trial vector $C = (1, c)$ and minimise it with respect to c . Show that the minimum is the lower root of the characteristic polynomial.
2. Expand E_- to second order in g and compare with the reference energy $E_{\text{ref}} = -g/2$ and with second-order perturbation theory.
3. What is c at the minimum, and what does its sign tell you about the ground state?

Answer to pause #3

1. $E(c) = [-\frac{g}{2} - gc + (2\xi - \frac{g}{2})c^2]/(1 + c^2)$. The condition $dE/dc = 0$ is the gradient formula, $HC = EC$ with $C = (1, c)$: $-\frac{g}{2} - \frac{g}{2}c = E$ and $-\frac{g}{2} + (2\xi - \frac{g}{2})c = Ec$. Eliminating c gives $(E + \frac{g}{2})(E - 2\xi + \frac{g}{2}) = \frac{g^2}{4}$, the characteristic polynomial; the minimum is its lower root $E_- = \xi - \frac{g}{2} - \sqrt{\xi^2 + g^2/4}$.
2. $\sqrt{\xi^2 + g^2/4} = \xi + g^2/(8\xi) + \mathcal{O}(g^4)$, so $E_- = -\frac{g}{2} - g^2/(8\xi) + \dots$: the reference energy plus the second-order term $|-g/2|^2/(0 - 2\xi) = -g^2/(8\xi)$. Diagonalisation sums the series to all orders.
3. $c = -2(E_- + \frac{g}{2})/g = (2/g)[\sqrt{\xi^2 + g^2/4} - \xi] > 0$ for $g > 0$ ($c = 0.236$ at $\xi = g = 1$): the excited pair enters *in phase* with the reference because the coupling is attractive; for $g < 0$ the sign flips. This c is tomorrow's amplitude C_{ij}^{ab} , and the t of the exponential ansatz.

The structure of the Hamiltonian matrix (section 5.4)

Six fermions below the Fermi level, determinants ordered by excitation level:

	$0p0h$	$1p1h$	$2p2h$	$3p3h$	$4p4h$	$5p5h$	$6p6h$
$0p0h$	E_{ref}	f_{ai}	$\langle ab \hat{v} ij\rangle$	0	0	0	0
$1p1h$	f_{ia}	$\times$	$\times$	$\times$	0	0	0
$2p2h$	$\langle ij \hat{v} ab\rangle$	$\times$	$\times$	$\times$	$\times$	0	0
$3p3h$	0	$\times$	$\times$	$\times$	$\times$	$\times$	0
$4p4h$	0	0	$\times$	$\times$	$\times$	$\times$	$\times$
$5p5h$	0	0	0	$\times$	$\times$	$\times$	$\times$
$6p6h$	0	0	0	0	$\times$	$\times$	$\times$

- The zeros are not accidental: a two-body operator connects determinants differing by at most two orbitals (Condon-Slater, week 36), and an $np-nh$ and an $mp-mh$ determinant differ by at least $|n-m|$ orbitals. The matrix is **banded in the excitation level, bandwidth two**, and overwhelmingly sparse.
- The first row and column are week 37's Example 3: E_{ref} , f_{ai} , $\langle ab|\hat{v}|ij\rangle$, then nothing. Section 4 of the notebook builds the 70×70 matrix of a *random* two-body Hamiltonian ($n=8$, $N=4$) and checks all 69 entries of the first column against these formulas, sign for sign.
- A three-body force widens the band to three and multiplies the number of non-zero elements per row by $\sim N(n-N)$ – which is why three-body forces are normal-ordered with respect to $|\Phi_0\rangle$ and truncated at the two-body level (section 3.17 of the notes).

The blocks in particle-hole language: every entry is a rule of week 36–37

Take the row of $\langle \Phi_i^a |$ and the row of $\langle \Phi_{ij}^{ab} |$, and ask what the generalised Wick theorem relative to $|\Phi_0\rangle$ allows:

element	value (antisymmetrised, $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$)	rule
$\langle \Phi_i^a \hat{H} \Phi_0 \rangle$	f_{ai}	one orbital, from $\hat{F}_N$ alone
$\langle \Phi_i^a \hat{H} \Phi_i^a \rangle$	$E_{\text{ref}} + f_{aa} - f_{ii} - \langle ai v ai \rangle$	diagonal (Example 4)
$\langle \Phi_i^a \hat{H} \Phi_j^b \rangle$	$\delta_{ij} f_{ab} - \delta_{ab} f_{ji} + \langle aj v ib \rangle$	one or two orbitals: the TDA matrix
$\langle \Phi_i^a \hat{H} \Phi_{jk}^{bc} \rangle$	$\delta_{ij} \langle ak v bc \rangle$ -type terms, $\pm f$ terms with two deltas	the $1p1h$ - $2p2h$ block
$\langle \Phi_i^a \hat{H} \Phi_{jkl}^{bcd} \rangle$	$\delta \delta \langle kl v cd \rangle$ -type terms only	the last non-zero block of the row
$\langle \Phi_{ij}^{ab} \hat{H} \Phi_0 \rangle$	$\langle ab v ij \rangle$	two orbitals, from $\hat{V}_N$ alone
$\langle \Phi_{ij}^{ab} \hat{H} \Phi_{ijkl}^{abcd} \rangle$	$\langle kl v cd \rangle$	two orbitals – the $2p2h$ - $4p4h$ block

- The band structure is the Condon-Slater table read row by row: $\hat{F}_N$ changes the excitation level by at most one, $\hat{V}_N$ by at most two, and each element is a product of Kronecker deltas (the orbitals the two determinants share) and one f or one v (the orbitals in which they differ).
- Tomorrow we write the $1p1h$ and $2p2h$ rows out in full, with the amplitudes C attached – the singles and doubles equations – and draw them.

In a Hartree-Fock basis: Brillouin's theorem empties one more block

If the single-particle states diagonalise the Fock matrix, $f_{ai} = 0$, and the reference does not couple to the singles at all:

	$0p0h$	$1p1h$	$2p2h$	$3p3h$	$4p4h$	$5p5h$	$6p6h$
$0p0h$	$\tilde{x}$	0	$\tilde{x}$	0	0	0	0
$1p1h$	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	0	0	0
$2p2h$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	0	0
$3p3h$	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	0
$4p4h$	0	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$
$5p5h$	0	0	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$
$6p6h$	0	0	0	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$

the tildes a reminder that the unitary transformation of the basis has changed every matrix element.

- The correlation energy is then carried entirely by the doubles at leading order – the starting point of both perturbation theory and coupled-cluster theory.
- For the pairing model *every* even-odd block is empty, whatever the basis: the interaction moves pairs and cannot break one. The symmetry does more than Brillouin's theorem would.
- Tomorrow afternoon: how to *find* the basis in which $f_{ai} = 0$. That is Hartree-Fock theory.

Pause to think: reading the block table

Pause to think #4

(about 3 min – discuss with your neighbour)

1. N particles in n orbitals. How many determinants sit at excitation level k ? Verify that the levels add up to $\binom{n}{N}$, and evaluate for $n = 12$, $N = 6$.
2. In the $3p3h$ row of the six-particle table, which blocks are non-zero, and which piece of $\hat{H}$ (E_{ref} , $\hat{F}_N$, $\hat{V}_N$) is responsible for each?
3. Suppose $\hat{H}$ also contains a three-body force $\hat{W}$. Which additional blocks fill up, and what happens to the cost of $\hat{H}|\Phi\rangle$ in the bit code?

Answer to pause #4

1. $\binom{N}{k} \binom{n-N}{k}$: choose which k occupied orbitals to empty and which k empty ones to fill. Vandermonde's identity gives $\sum_k \binom{N}{k} \binom{n-N}{k} = \binom{n}{N}$. For $n = 12$, $N = 6$: $\binom{6}{k}^2 = 1, 36, 225, 400, 225, 36, 1$, total 924 – this evening's six-level pairing model.
2. Non-zero: $1p1h$ to $5p5h$. The diagonal block gets all three pieces; the neighbours $2p2h$ and $4p4h$ get $\hat{F}_N$ (moves one orbital) and $\hat{V}_N$ (moves one orbital with a spectator line – the interaction with the *excited* particles and holes, which f does not contain); $1p1h$ and $5p5h$ get $\hat{V}_N$ only.
3. Bandwidth three: the $0p0h$ – $3p3h$, $1p1h$ – $4p4h$, ... blocks fill. The three-body loop of the bit code runs over three occupied and three arbitrary orbitals, $\mathcal{O}(N^3 n^3)$ against $\mathcal{O}(N^2 n^2)$ per determinant – a factor ~ 320 for $n = 40$, $N = 8$ – and $\langle pqr|w|stu \rangle$ needs n^6 storage. Hence the normal-ordered two-body approximation of section 3.17.

Truncating the expansion (section 5.8)

Since the full expansion is out of reach for all but the smallest systems, we truncate it by excitation level:

$$|\Psi_{\text{CIS}}\rangle = (1 + \hat{C}_1)|\Phi_0\rangle, \quad |\Psi_{\text{CID}}\rangle = (1 + \hat{C}_2)|\Phi_0\rangle, \quad |\Psi_{\text{CISD}}\rangle = (1 + \hat{C}_1 + \hat{C}_2)|\Phi_0\rangle, \dots$$

up to CISDTQ and, at $\hat{C}_N$, full CI. Each is the Ritz method in a subspace: an upper bound to E_0 , decreasing monotonically as the truncation is relaxed (the interlacing theorem of lecture 2). The space at level k holds $\binom{N}{k} \binom{n-N}{k}$ determinants, polynomial of degree $2k$, and the work grows roughly as n^{2k+2} : CISD at $\mathcal{O}(n^6)$ is affordable, CISDTQ at $\mathcal{O}(n^{10})$ is not.

g	E_{ref}	CIS (17)	CID (37)	CISD (53)	FCI (70)
0.25	1.750000	1.75000000	1.73050700	1.73050700	1.73045985
0.50	1.500000	1.50000000	1.41759645	1.41759645	1.41677428
1.00	1.000000	1.00000000	0.64890655	0.64890655	0.63554847
2.00	0.000000	0.00000000	-1.35208670	-1.35208670	-1.48965216

- The singles change nothing (a single excitation breaks a pair – the reference is decoupled from the whole singles block), and CISD coincides with CID.
- The doubles carry 99.8% of the correlation energy at $g = 0.25$, 96.3% at $g = 1$, only 90.8% at $g = 2$: truncation at fixed level works while the reference is good and degrades as the coupling grows.

Six levels, six particles: 924 determinants (Figure 5.2 of the notes)

A space large enough for the higher truncations to mean something: $\binom{12}{6} = 924$ determinants, 1, 36, 225, 400, 225, 36, 1 over the levels; CIS keeps 37, CISD 262, CISDTQ 887 (section 5 of the notebook, a few seconds with the bit code):

g	E_{ref}	CISD	CISDTQ	FCI	CISD, % of corr.	CISDTQ, % of corr.
0.05	5.925	5.92381898	5.92381873	5.92381873	99.98	100.0000
0.25	5.625	5.59295380	5.59276099	5.59276086	99.40	99.9996
0.50	5.250	5.10969074	5.10593698	5.10592638	97.39	99.9926
1.00	4.500	3.87227845	3.80238641	3.80152797	89.87	99.8771
2.00	3.000	0.54353367	-0.15967891	-0.18915925	77.03	99.0756

- CIS stays at E_{ref} at every g (not shown). At $g = 0.05$ the doubles are wrong by 2.5×10^{-7} and the quadruples by 6.5×10^{-12} ; at $g = 2$ by 0.73 and 2.9×10^{-2} : the error of a *fixed* truncation grows by six to ten orders of magnitude while nothing changes but the coupling.
- The gain from adding a level shrinks as the problem hardens: CISDTQ over CISD by 4×10^4 at $g = 0.05$, by 82 at $g = 1$, by 25 at $g = 2$ – the correction is largest exactly where its estimate is worst.
- At $g = 2$ CISD recovered 90.8% with four levels and recovers 77.0% with six: **the same truncation gets worse as the system grows**. That is the next slide, seen from the side.

Size consistency (section 5.9)

There is a defect of truncated CI more serious than the loss of accuracy, because it does not go away as the basis is improved. Two identical systems A and B , infinitely far apart: $\hat{H} = \hat{H}_A + \hat{H}_B$, the exact ground state is the product $|\Psi_A\rangle|\Psi_B\rangle$, and

$$E(AB) = E(A) + E(B) = 2E(A) \quad (\text{size consistency}).$$

Full CI passes, since the product state lies in the full space. Truncated CI does not: if A and B are each doubly excited, the combined state is a *quadruple* excitation of the combined reference, and CID has no room for it. The product of two doubles is a quadruple – truncation at a fixed level is not multiplicatively closed.

g	$2E(A)$	$E(AB)$, FCI	$E(AB)$, CID	CID error	$E(AB)$, CISDTQ
0.25	-0.26556444	-0.26556444	-0.26550480	5.96×10^{-5}	-0.26556444
0.50	-0.56155281	-0.56155281	-0.56066017	8.93×10^{-4}	-0.56155281
1.00	-1.23606798	-1.23606798	-1.22474487	1.13×10^{-2}	-1.23606798

Two non-interacting copies of the two-level, one-pair system of pause #3 ($E(A) = E_-$, section 5 of the notebook). CISDTQ is exact for two copies – the missing quadruple is back – and fails again for three (Figure 5.3 of the notes: error 9.1×10^{-4} at $g = 1$). With M subsystems the error grows with M : the energy per particle of truncated CI drifts with the size of the system – fatal for nuclear matter, the electron gas, a large molecule.

The repair: make the ansatz multiplicative

For two non-interacting one-pair systems the exact state is a product,

$$(1 + c\hat{A}_2^A)(1 + c\hat{A}_2^B)|\Phi_0\rangle = (1 + c\hat{A}_2^A + c\hat{A}_2^B + c^2\hat{A}_2^A\hat{A}_2^B)|\Phi_0\rangle,$$

and the last term – both subsystems excited at once – is a $4p4h$ excitation of the combined reference, absent from CID, with a coefficient that is *not* independent: c^2 , the product of the two doubles.

Coupled-cluster theory in one line

$$|\Psi_{\text{CCSD}}\rangle = e^{\hat{T}_1 + \hat{T}_2}|\Phi_0\rangle = (1 + \hat{T}_1 + \hat{T}_2 + \frac{1}{2}\hat{T}_2^2 + \dots)|\Phi_0\rangle$$

The quadruple $\frac{1}{2}\hat{T}_2^2$ appears automatically with its coefficient fixed by the doubles; for non-interacting subsystems $e^{\hat{T}} = e^{\hat{T}_A}e^{\hat{T}_B}$ and size consistency is exact – with the *same* number of parameters as CISD.

- Section 5 of the notebook does it with *one* parameter: the vector $(1, t, t, t^2)$ on the four paired configurations of the two copies, minimised over t , gives $E = -1.23606797$ at $t = 0.2361$ – the FCI answer, and $t = c$ of pause #3.
- This is why coupled cluster, not truncated CI, is the workhorse of modern many-body physics; we develop it properly later in the course. Tomorrow's diagrams show *where* in the CI equations the defect sits.

Summary of Thursday, and what comes on Friday

Today

- Bit patterns: a determinant is an integer, $\hat{H}|\Phi\rangle$ a loop with a popcount
- Pairing model: 70 = 6 to every digit; $-\frac{7}{24}g^2$ at leading order
- Ritz = optimise the coefficients; multiplier = energy; $\det(H - E1) = 0$ defines, never computes; interlacing bounds
- Banded matrix from Condon-Slater; first row $E_{\text{ref}}, f_{ai}, \langle ab|v|ij\rangle, 0$; HF basis empties $0p0h-1p1h$
- Truncation: 99.8% $\rightarrow$ 77%; not size consistent; $e^{\hat{T}}$ is

Before Friday: run sections 1–5 of `week38.ipynb`; redo pause #3 for $g < 0$ and check the sign of c .

Friday

- The FCI equation projected: energy, singles and doubles equations, every term; the same equations as **diagrams**
- The non-practical way: iterate the amplitude equations; first iterate = second-order perturbation theory; what each method keeps
- Hartree-Fock theory, chapter 6: mean field, variational calculus, the HF equations, the self-consistent field, Brillouin

Friday, lecture 3: The FCI equations projected, drawn and iterated

The FCI equation as coupled equations for the amplitudes (section 5.10)

With intermediate normalisation, $|\Psi_0\rangle = (1 + \hat{C}_1 + \hat{C}_2 + \hat{C}_3 + \dots)|\Phi_0\rangle$, antisymmetric amplitudes and unrestricted sums,

$$\hat{C}_1 = \sum_{ai} C_i^a a_a^\dagger a_i, \quad \hat{C}_2 = \frac{1}{4} \sum_{abij} C_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i, \quad \hat{C}_3 = \frac{1}{36} \sum_{abij} C_{ijk}^{abc} a_a^\dagger a_b^\dagger a_c^\dagger a_k a_j a_i, \dots$$

project $(\hat{H} - E)|\Psi_0\rangle = 0$ onto the reference and onto each excited determinant:

$$\begin{aligned} \langle \Phi_0 | \hat{H} | \Psi_0 \rangle &= E \langle \Phi_0 | \Psi_0 \rangle = E, & \text{the energy equation,} \\ \langle \Phi_i^a | \hat{H} | \Psi_0 \rangle &= E C_i^a, & \text{the singles equations,} \\ \langle \Phi_{ij}^{ab} | \hat{H} | \Psi_0 \rangle &= E C_{ij}^{ab}, & \text{the doubles equations, ...} \end{aligned}$$

- Exactly the rows of $HC = EC$ with $C_0 = 1$. Nothing new – but with $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$ and the generalised Wick theorem relative to $|\Phi_0\rangle$, each row becomes a short list of terms, one per way of contracting a piece of $\hat{H}$ with a piece of $\hat{C}$.
- Yesterday's band structure as a rule for the *equations*: $\hat{F}_N$ shifts the excitation level by at most one, $\hat{V}_N$ by at most two, so the singles equation contains $\hat{C}_0, \dots, \hat{C}_3$ and the doubles equation $\hat{C}_0, \dots, \hat{C}_4$.

The energy equation

$\langle \Phi_0 | (E_{\text{ref}} + \hat{F}_N + \hat{V}_N) (1 + \hat{C}_1 + \hat{C}_2 + \dots) | \Phi_0 \rangle$: the normal-ordered pieces have zero expectation value, $\hat{F}_N$ can only close on $\hat{C}_1$ (two lines) and $\hat{V}_N$ only on $\hat{C}_2$ (four lines):

The correlation energy

$$\Delta E \equiv E - E_{\text{ref}} = \sum_{ia} f_{ia} C_i^a + \frac{1}{4} \sum_{ijab} \langle ij | \hat{v} | ab \rangle C_{ij}^{ab} \quad \left(\frac{1}{4} \rightarrow \sum_{i < j, a < b} \text{ with restricted sums} \right).$$

- The correlation energy is fixed by the **singles and doubles amplitudes alone**, whatever the excitation level of the expansion – the reference has no matrix element with anything beyond the doubles. The triples, quadruples, ... enter only *through* C_i^a and C_{ij}^{ab} , by the coupled equations of the next slides.
- In a Hartree-Fock basis $f_{ia} = 0$ and $\Delta E = \frac{1}{4} \sum \langle ij | \hat{v} | ab \rangle C_{ij}^{ab}$: doubles only. This is the equation of `fci.do.txt` and the reason the doubles dominate every method of the course.
- Section 6 of the notebook: for the random $n = 8$, $N = 4$ Hamiltonian, $\Delta E = -5.7494604184$ from the eigenvalue and -5.7494604184 from the formula, with the exact eigenvector; -4.85 of it from the singles term in the original basis, 0 from the singles in the HF basis (section 8).

The singles equations, all eight terms

$\langle \Phi_i^a | = \langle \Phi_0 | a_i^\dagger a_a$ supplies two quasiparticle annihilators; contract them, together with $\hat{F}_N$ or $\hat{V}_N$, into the amplitude vertices (generalised Wick theorem relative to $|\Phi_0\rangle$; the notebook, section 6, evaluates every term with the FCI eigenvector and verifies the equation to 10^{-14} for all 16 pairs (a, i)):

$$\langle \Phi_i^a | \hat{H} | \Psi_0 \rangle = E C_i^a, \text{ i.e.}$$

$$\begin{aligned} \Delta E C_i^a = & \underbrace{f_{ai}}_{\hat{F}_N \cdot 1} + \underbrace{\sum_b f_{ab} C_i^b - \sum_j f_{ji} C_j^a}_{\hat{F}_N \cdot \hat{C}_1} + \underbrace{\sum_{jb} \langle aj | \hat{v} | ib \rangle C_j^b}_{\hat{V}_N \cdot \hat{C}_1} + \underbrace{\sum_{jb} f_{jb} C_{ij}^{ab}}_{\hat{F}_N \cdot \hat{C}_2} \\ & + \underbrace{\frac{1}{2} \sum_{jbc} \langle aj | \hat{v} | bc \rangle C_{ij}^{bc} - \frac{1}{2} \sum_{jkb} \langle jk | \hat{v} | ib \rangle C_{jk}^{ab}}_{\hat{V}_N \cdot \hat{C}_2} + \underbrace{\frac{1}{4} \sum_{jkb} \langle jk | \hat{v} | bc \rangle C_{ijk}^{abc}}_{\hat{V}_N \cdot \hat{C}_3}. \end{aligned}$$

- The E_{ref} piece gives $E_{\text{ref}} C_i^a$, moved to the left: $\Delta E C_i^a$. Each term on the right is one matrix element times one amplitude – **linear** in the C 's, but ΔE depends on them through the energy equation.
- Counting: two lines from the bra, two from $\hat{F}_N$ or four from $\hat{V}_N$, $2n$ from $\hat{C}_n$, every line contracted, none inside a group; $\hat{F}_N \hat{C}_3$ and $\hat{V}_N \hat{C}_4$ leave lines over. The band of yesterday.

The doubles equations

Same procedure with $\langle \Phi_{ij}^{ab} | = \langle \Phi_0 | a_i^\dagger a_j^\dagger a_b a_a$ (four annihilators; $P(ab)X_{ab} = X_{ab} - X_{ba}$; verified in the notebook for all 36 pairs to 10^{-14}):

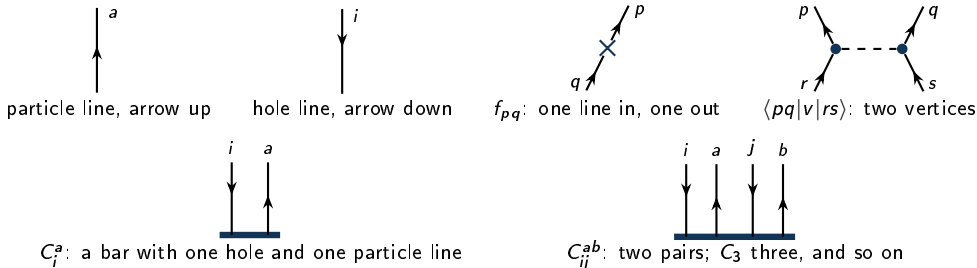
$$\langle \Phi_{ij}^{ab} | \hat{H} | \Psi_0 \rangle = E C_{ij}^{ab}$$

$$\begin{aligned} \Delta E C_{ij}^{ab} = & \langle ab | \hat{v} | ij \rangle + P(ab) \sum_c f_{bc} C_{ij}^{ac} - P(ij) \sum_k f_{kj} C_{ik}^{ab} + \frac{1}{2} \sum_{cd} \langle ab | \hat{v} | cd \rangle C_{ij}^{cd} + \frac{1}{2} \sum_{kl} \langle kl | \hat{v} | ij \rangle C_{kl}^{ab} \\ & + P(ij) P(ab) \sum_{kc} \langle kb | \hat{v} | cj \rangle C_{ik}^{ac} \\ & + P(ij) \sum_c \langle ab | \hat{v} | cj \rangle C_i^c - P(ab) \sum_k \langle kb | \hat{v} | ij \rangle C_k^a + P(ij) P(ab) f_{ai} C_j^b \\ & + \sum_{kc} f_{kc} C_{ijk}^{abc} + \frac{1}{2} P(ab) \sum_{kcd} \langle bk | \hat{v} | cd \rangle C_{ijk}^{acd} - \frac{1}{2} P(ij) \sum_{klc} \langle kl | \hat{v} | jc \rangle C_{ikl}^{abc} + \frac{1}{4} \sum_{klcd} \langle kl | \hat{v} | cd \rangle C_{ijkl}^{abcd}. \end{aligned}$$

- First two lines: the driving term and the five **doubles-doubles** terms – the whole of CID, and the linear part of coupled-cluster doubles. Third line: singles. Last line: triples and quadruples.
- The red term is **disconnected**: f_{ai} times an amplitude sharing no index with it, the product of two lower pieces (zero in a Hartree-Fock basis). With $\Delta E C_{ij}^{ab}$ on the left it is the algebraic fingerprint of yesterday's size-consistency failure; the exponential ansatz cancels both against the $\frac{1}{2} \hat{T}_2^2$ terms (chapter 8).

Drawing the equations: the elements

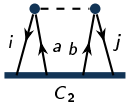
Goldstone-style diagrams, time running *upward*; $|\Phi_0\rangle$ is the vacuum and is not drawn.



- An amplitude C_n is a bar from which n particle-hole pairs emerge; $\hat{V}_N$ a dashed line joining two vertices, each with one line in and one out, read as $\langle \text{out}_1 \text{out}_2 | v | \text{in}_1 \text{in}_2 \rangle$ (antisymmetrised); $\hat{F}_N$ a cross, $f_{\text{out},\text{in}}$.
- The bra $\langle \Phi_H^P |$ fixes the **open lines at the top**: none for the energy equation, one pair (a, i) for the singles, two pairs for the doubles; every other line is internal and summed over. A term of the equations is one way of joining the open lines, one piece of $\hat{H}$ and one bar with *every* line of $\hat{H}$ and of the bar connected – the generalised Wick theorem, drawn.

The energy equation, drawn

$$\Delta E = \text{diagram } C_1 + \text{diagram } C_2 = \sum_{ia} f_{ia} C_i^a + \frac{1}{4} \sum_{ijab} \langle ij | \hat{v} | ab \rangle C_{ij}^{ab}.$$



Rules for the value of a diagram

1. Sum over all internal labels; write $f_{\text{out},\text{in}}$ for every cross, $\langle \text{out out} | v | \text{in in} \rangle$ for every interaction, and the amplitude of every bar.
2. A factor $\frac{1}{2}$ for each pair of **equivalent lines**: two lines of the same kind joining the same two elements in the same direction.
3. A sign $(-1)^{h+l}$, h the number of hole lines, l the number of closed loops – for open diagrams counting the *quasi-loops* obtained by joining each open particle line to its partner hole line at the top.
4. For open diagrams, a permutation operator P for each pair of open lines that are *not* equivalent.

First diagram: $h = 1$, $l = 1$, no equivalent lines: $+f_{ia} C_i^a$. Second: $h = 2$, $l = 2$, two equivalent pairs (i, j and a, b): $+\frac{1}{4} \langle ij | v | ab \rangle C_{ij}^{ab}$. ✓

The singles equation, drawn: the $\hat{C}_1$ terms

Open lines a (up) and i (down) at the top; $\Delta E C_i^a =$

$$f_{ai} + \sum_b f_{ab} C_i^b - \sum_j f_{ji} C_j^a + \sum_{jb} \langle aj | v | ib \rangle C_j^b + \dots$$

- Diagram 1: the driving term, $h = 1$, $l = 1$ (the quasi-loop $a-i$): $+f_{ai}$. Diagram 2: the cross on the particle line, $h = 1$, $l = 1$: $+f_{ab} C_i^b$. Diagram 3: the cross on the hole line, $h = 2$ (i and j), $l = 1$: $-f_{ji} C_j^a$. Diagram 4: the bar closes a *bubble* (j, b) on one vertex, the other vertex carries the open lines: $h = 2$, $l = 2$: $+\langle aj | v | ib \rangle C_j^b$.
- In a Hartree-Fock basis diagram 1 is absent and diagrams 2, 3 collapse to $(\epsilon_a - \epsilon_i) C_i^a$: the “energy denominator” of perturbation theory is the crosses on the open lines.

The singles equation, drawn: the $\hat{C}_2$ and $\hat{C}_3$ terms

$$\begin{aligned}
 & \dots + \text{Diagram 5} + \text{Diagram 6} + \text{Diagram 7} + \text{Diagram 8} \\
 & + \sum_{jb} f_{jb} C_{ij}^{ab} + \frac{1}{2} \sum_{jbc} \langle aj | v | bc \rangle C_{ij}^{bc} - \frac{1}{2} \sum_{jkb} \langle jk | v | ib \rangle C_{jk}^{ab} + \frac{1}{4} \sum_{jkb} \langle jk | v | bc \rangle C_{ijk}^{abc}
 \end{aligned}$$

- Diagram 5: a bubble on the cross, $h = 2$, $l = 2$: +. Diagram 6: two particle lines b, c from the bar into the interaction, equivalent $(\frac{1}{2})$; $h = 2$, $l = 2$: +. Diagram 7: its mirror image with two equivalent hole lines; $h = 3$, $l = 2$: -. Diagram 8: two bubbles on the two vertices, two equivalent pairs $(\frac{1}{4})$; $h = 3$, $l = 3$: +.
- Every diagram is **connected** to the open lines: in the singles equation there is no room for a disconnected piece. That changes for the doubles.

The doubles equation, drawn

Open lines a, b (up) and i, j (down); the driving term, the five doubles-doubles diagrams, and the disconnected singles term:

$$\begin{aligned}
 &\langle ab|v|ij\rangle & P(ab) \sum_c f_{bc} C_{ij}^{ac} & -P(ij) \sum_k f_{kj} C_{ik}^{ab} & \frac{1}{2} \sum_{cd} \langle ab|v|cd\rangle C_{ij}^{cd} \\
 &\frac{1}{2} \sum_{kl} \langle kl|v|ij\rangle C_{kl}^{ab} & P(ij)P(ab) \sum_{kc} \langle kb|v|cj\rangle C_{ik}^{ac} & P(ij)P(ab) f_{ai} C_j^b
 \end{aligned}$$

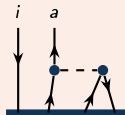
- Crosses on an open line, particle and hole ladders ($\frac{1}{2}$ per equivalent pair), a ring with four inequivalent open lines ($P(ij)P(ab)$): with the driving term, this is CID – and the linear part of coupled cluster.
- The last diagram is **two pieces**, the open f_{ai} vertex beside a C_1 bar: disconnected. The linked-cluster theorem removes such terms; CI keeps them and pays with size consistency.

Pause to think: from diagram to term and back

Pause to think #5

(about 4 min – discuss with your neighbour)

1. Write down the value of the diagram below (open lines a , i) using the four rules, and identify it in the singles equation.



2. Draw the diagram for $-\frac{1}{2}P(ij) \sum_{klc} \langle kl|\hat{v}|jc\rangle C_{ikl}^{abc}$ in the doubles equation and check its sign and factor with the rules.
3. Which diagrams of the singles equation survive if the amplitudes are those of CID ($C_1 = C_3 = 0$)? What does the singles equation then say?

Answer to pause #5

1. A C_2 bar with i open, particles b, c into the two vertices, hole j from the right vertex back to the bar, a emitted by the left vertex: $\langle aj|v|bc\rangle C_{ij}^{bc}$ summed over j, b, c ; b, c equivalent: $\frac{1}{2}$; $h = 2$, loops $b \rightarrow a \rightarrow i \rightarrow \text{bar}$ and $c \rightarrow j \rightarrow \text{bar}$: $l = 2$, sign $+$. Diagram 6.
2. A C_3 bar with open a, b, i ; holes k, l from the bar into the two vertices; one vertex re-emits the open hole j , the other absorbs the particle c . Equivalent k, l : $\frac{1}{2}$; open holes i, j inequivalent: $P(ij)$; $h = 4$, loops $a-i, b-j$ (through l), $k-c$: $l = 3$, sign $(-1)^7 = -1$. ✓
3. Diagrams 1, 5, 6, 7 – four terms and no C_i^a on the left; $f_{ai} + \sum f_{jb} C_{ij}^{ab} + \dots \neq 0$ in general. CID does not solve the singles rows, it *drops* them: Ritz in a subspace projects onto the subspace and satisfies only its own rows.

A non-practical way of solving the FCI equations (fci.do.txt, section 5.10)

Take the singles equation and isolate the diagonal term: $\langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle = E_{\text{ref}} + f_{aa} - f_{ii} - \langle ai | v | ai \rangle \equiv H_{ia,ia}$, so that

$$(E - H_{ia,ia}) C_i^a = f_{ai} + \sum_{bj \neq ai} \langle \Phi_i^a | \hat{H} | \Phi_j^b \rangle C_j^b + \sum_{bcjk} \langle \Phi_i^a | \hat{H} | \Phi_{jk}^{bc} \rangle C_{jk}^{bc} + \sum_{bcdjkl} \langle \Phi_i^a | \hat{H} | \Phi_{jkl}^{bcd} \rangle C_{jkl}^{bcd},$$

and likewise for every other row. A prescription:

Iteration

1. Guess the amplitudes – from first-order perturbation theory, $C_i^a = f_{ai} / (\epsilon_i - \epsilon_a)$, $C_{ij}^{ab} = \langle ab | v | ij \rangle / (\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b)$.
 2. Compute E from the energy equation, $E = E_{\text{ref}} + \sum f_{ia} C_i^a + \frac{1}{4} \sum \langle ij | v | ab \rangle C_{ij}^{ab}$.
 3. Compute the right-hand side of every row with the current amplitudes, divide by $E - H_{K,K}$, update every C_K ; repeat from 2 until nothing moves.
- The right-hand sides are the diagrams of the previous slides; E on the left makes the scheme **non-linear** in the amplitudes although each row is linear. Pairing model, four levels (section 7 of the notebook): the **first iterate is exactly** $E_{\text{ref}} - \frac{7}{24} g^2$, the second-order energy of pause #2; at $g = 1$ the sequence 0.7083, 0.6359 (5 iterations), 0.63554847 (20) reaches the FCI value.

What the iteration is, and what it is not

- A Jacobi iteration on $HC = EC$ with the energy updated on the way. It converges when the reference dominates the ground state, and it produces, order by order, the **perturbation series**: the m -th iterate contains the right-hand side substituted into itself m times – chapter 7 does the bookkeeping, with the same diagrams and energy denominators from the crosses on the open lines.
- *Not* a way around the wall: the loop runs over all $\binom{n}{N}$ amplitudes and every right-hand side costs as much as $\hat{H}|\Phi\rangle$. Lanczos does the same work far more cleverly.
- It *is* the template for every practical method – each a rule for which rows and diagrams to keep, and how to close the system:

method	amplitudes	how the equations are closed
Hartree-Fock	none: C_0 only	vary the <i>orbitals</i> until $f_{ai} = 0$
perturbation theory	all, order by order	a fixed number of iterations, $E \rightarrow E_{\text{ref}}$ in the denominators
Tamm-Dancoff, RPA	singles (and ring doubles)	diagonalise the $1p1h$ block; sum the rings
truncated CI	up to a fixed level	Ritz in the subspace: linear, unlinked terms kept
coupled cluster	up to a fixed level	$e^{\hat{T}}$: same diagrams plus non-linear ones; unlinked terms cancel

Full CI is the fixed point that measures each answer: whenever a model is small enough to solve exactly, solve it and calibrate.

The link to Hartree-Fock: kill the driving term

Look at the singles equation once more. Its driving term is f_{ai} , the same quantity that couples the reference to the singles in the energy equation:

$$\Delta E = \sum_{ia} f_{ia} C_i^a + \frac{1}{4} \sum \langle ij|v|ab \rangle C_{ij}^{ab}, \quad \Delta E C_i^a = f_{ai} + \sum_b f_{ab} C_i^b - \sum_j f_{ji} C_j^a + \dots$$

- If the single-particle basis could be chosen such that $f_{ai} = 0$ for all a, i , the singles would be driven by the doubles only, the energy equation would contain the doubles alone, and the reference would be the best single determinant *in the sense of the variational principle*.
- That choice exists: the Fock matrix $f_{pq} = h_{pq} + \sum_i \langle pi|v|qi \rangle$ depends on the occupied orbitals, and diagonalising it self-consistently – the occupied orbitals being its own lowest eigenvectors – makes $f_{ai} = 0$. This is the **Hartree-Fock** basis, and $f_{ai} = 0$ is **Brillouin's theorem**.
- Section 8 of the notebook, random Hamiltonian of yesterday: $E_{\text{ref}} = 6.497$, $E^{\text{HF}} = 1.298$ after 48 iterations, $E_0^{\text{FCI}} = 0.748$ in *both* bases; in the HF basis $\max |f_{ai}| = 9 \times 10^{-11}$, the $0p0h-1p1h$ block is empty and $\Delta E = -0.550$ comes from the doubles term alone.

Many-body connection

Chapter 5 optimised the *expansion* on a fixed reference. Chapter 6 keeps one determinant and optimises the *reference*. Both are the variational principle; the second is the last lecture of this week.

Summary of lecture 3

- Projecting $(\hat{H} - E)|\Psi_0\rangle = 0$ row by row gives the energy equation (ΔE from singles and doubles only), the singles equation (eight terms, C_0 to C_3) and the doubles equation (thirteen terms, C_0 to C_4) – every term a product of one f or v and one amplitude, all verified numerically in the notebook.
- The same equations are diagrams: bars for amplitudes, crosses and dashed lines for $\hat{F}_N$ and $\hat{V}_N$, open lines from the bra; value by four rules – sums, $\frac{1}{2}$ per equivalent pair, $(-1)^{h+l}$, permutations. The doubles equation contains a **disconnected** diagram, the fingerprint of the size-consistency failure.
- Iterating the rows from a perturbative guess reproduces second-order perturbation theory at the first step and converges to FCI; every practical method is a rule for which rows and which diagrams to keep.
- Making the driving term f_{ai} vanish by choosing the orbitals is Hartree-Fock theory – next.

Friday, lecture 4: Hartree-Fock theory

Why a mean field? (section 6.1)

Chapter 5 showed that the exact answer is a superposition of exponentially many determinants and that we cannot afford it. Chapter 6 takes the opposite tack: **keep exactly one determinant, and ask which one is best.**

Hartree-Fock theory in two ingredients

A single-particle basis defined by its own eigenvalue problem,

$$\hat{h}^{\text{HF}} \psi_{\alpha} = \varepsilon_{\alpha} \psi_{\alpha}, \quad \hat{h}^{\text{HF}} = \hat{t} + \hat{u}_{\text{ext}} + \hat{u}^{\text{HF}},$$

and the prescription for the still unknown one-body potential $\hat{u}^{\text{HF}}$: choose it such that $E^{\text{HF}} = \langle \Phi_0 | \hat{H} | \Phi_0 \rangle$ is stationary, with $|\Phi_0\rangle$ the determinant of the N lowest solutions.

- The variational principle guarantees $E^{\text{HF}} \geq E_0$: an upper bound, like every Ritz value of this week – here with the *orbitals* as parameters and the expansion frozen at one term.
- We shall find that $\hat{h}^{\text{HF}}$ is exactly the operator $\hat{f}$ of the normal-ordered Hamiltonian,

$$\langle p | \hat{h}^{\text{HF}} | q \rangle = \langle p | \hat{t} + \hat{u}_{\text{ext}} | q \rangle + \sum_{i \leq F} \langle p | \hat{v} | q i \rangle, \quad \langle p | \hat{u}^{\text{HF}} | q \rangle = \sum_{i \leq F} \langle p | \hat{v} | q i \rangle :$$

the interaction summed over the other fermions is an effective one-body field in which a given fermion moves – dependent on which states are occupied (**medium dependence**) and on the interaction. For atoms and molecules $\hat{u}_{\text{ext}}$ is the attraction of the nuclei; a nucleus has none, and the whole field must be generated self-consistently.

Variational calculus and Lagrange multipliers (section 6.2)

The same machinery as Thursday, in its simplest setting: one particle, $E = \int \psi^* \hat{H} \psi d^3r$ with $\int \psi^* \psi d^3r = 1$ and $\hat{H} = -\frac{1}{2} \nabla^2 + V$. Integrating the kinetic term by parts, the constrained variation is

$$\delta \left\{ \int \left(\frac{1}{2} \nabla \psi^* \cdot \nabla \psi + V \psi^* \psi - \lambda \psi^* \psi \right) d^3r \right\} = 0.$$

With $f = \frac{1}{2}(\psi_x^* \psi_x + \psi_y^* \psi_y + \psi_z^* \psi_z) + V \psi^* \psi - \lambda \psi^* \psi$, the Euler-Lagrange equation for ψ^* ,

$$\frac{\partial f}{\partial \psi^*} - \frac{\partial}{\partial x} \frac{\partial f}{\partial \psi_x^*} - \frac{\partial}{\partial y} \frac{\partial f}{\partial \psi_y^*} - \frac{\partial}{\partial z} \frac{\partial f}{\partial \psi_z^*} = 0 \quad \implies \quad -\frac{1}{2} \nabla^2 \psi + V \psi = \lambda \psi.$$

- The multiplier is the energy and the stationarity condition is the Schrödinger equation. **The pattern of Thursday:** a constrained variation produces an eigenvalue problem, and the multiplier enforcing normalisation is the eigenvalue. In FCI the unknowns were the coefficients C_H^P and the constraint $\sum |C|^2 = 1$; here the unknown is a function and the constraint an integral.
- In Hartree-Fock the unknowns are N orbitals, the constraints are the $N(N+1)/2$ orthonormality integrals $\langle i|j \rangle = \delta_{ij}$, and there will be one multiplier per constraint – reduced to one per orbital by the freedom of the next slide.

Determinants under a linear change of basis (section 6.3)

A determinant is linear in each column separately,

$$\begin{vmatrix} \alpha_1 b_{11} + \alpha_2 b_{12} & a_{12} \\ \alpha_1 b_{21} + \alpha_2 b_{22} & a_{22} \end{vmatrix} = \alpha_1 \begin{vmatrix} b_{11} & a_{12} \\ b_{21} & a_{22} \end{vmatrix} + \alpha_2 \begin{vmatrix} b_{12} & a_{12} \\ b_{22} & a_{22} \end{vmatrix}, \quad \text{and in general } \det(BC) = \det B \det C.$$

Expand new orbitals in a fixed orthonormal basis, $\psi_p(x) = \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x)$. The Slater determinant of the ψ 's is

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x_1) & \cdots & \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x_N) \\ \vdots & & \vdots \\ \sum_{\lambda} C_{t\lambda} \phi_{\lambda}(x_1) & \cdots & \sum_{\lambda} C_{t\lambda} \phi_{\lambda}(x_N) \end{vmatrix} = \det(C) \Phi(\phi),$$

with $\Phi(\phi)$ the determinant of the original basis functions. Two consequences, used constantly:

- **Orthonormality is preserved by a unitary** C : $w_j^{\dagger} w_i = v_j^{\dagger} U^{\dagger} U v_i = \delta_{ij}$. A unitary change of coefficients never spoils the basis.
- **A determinant is invariant, up to a phase, under any unitary mixing of its occupied orbitals**: it knows only which N -dimensional subspace they span. The Hartree-Fock solution is a statement about a **subspace**; the individual orbitals are fixed only by the convention that the Fock matrix be diagonal – which is what lets us use one multiplier per orbital.

The energy functional in terms of the coefficients (section 6.4)

The energy of a single determinant is the functional of week 34, E_{ref} of week 37:

$$E[\Phi] = \sum_{\mu=1}^N \langle \mu | \hat{h}_0 | \mu \rangle + \frac{1}{2} \sum_{\mu\nu=1}^N \langle \mu\nu | \hat{v} | \mu\nu \rangle_{\text{AS}}.$$

Insert $\psi_i = \sum_{\alpha} C_{i\alpha} \phi_{\alpha}$ (Greek indices over the whole fixed basis, Latin over the N occupied orbitals):

E as a function of the coefficients

$$E[\Phi^{\text{HF}}] = \sum_{i=1}^N \sum_{\alpha\beta} C_{i\alpha}^* C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle + \frac{1}{2} \sum_{i,j=1}^N \sum_{\alpha\beta\gamma\delta} C_{i\alpha}^* C_{j\beta}^* C_{i\gamma} C_{j\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{\text{AS}}.$$

- The coefficients are not free: $\langle i | j \rangle = \delta_{ij} = \sum_{\alpha} C_{i\alpha}^* C_{j\alpha}$. Introduce one multiplier ϵ_i per occupied orbital and make

$$F[\Phi^{\text{HF}}] = E[\Phi^{\text{HF}}] - \sum_{i=1}^N \epsilon_i \sum_{\alpha} C_{i\alpha}^* C_{i\alpha}$$

stationary. As on Thursday, C and C^* may be varied independently: $\partial F / \partial C_{i\alpha}^* = 0$.

- Compare with FCI: there E was *quadratic* in the unknowns and the stationarity condition linear (an eigenvalue problem). Here E is *quartic* in the C 's, and the condition will be an eigenvalue problem whose matrix depends on its own solution.

Carrying out the variation

Differentiate with respect to $C_{i\alpha}^*$. The one-body term gives $\sum_{\beta} C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle$. In the two-body term $C_{i\alpha}^*$ appears twice – as $C_{i\alpha}^*$ and as $C_{j\beta}^*$ with $j = i$ – with equal contributions by the symmetry of $\langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{AS}$ under the simultaneous interchange of both pairs; they cancel the $\frac{1}{2}$:

$$\sum_{\beta} C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{j=1}^N \sum_{\beta\gamma\delta} C_{j\beta}^* C_{j\delta} C_{i\gamma} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{AS} = \epsilon_i^{\text{HF}} C_{i\alpha}.$$

Relabel the dummy indices so that $C_{i\beta}$ factors out of both terms,
 $\sum_{\beta} \{ \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{j\gamma\delta} C_{j\gamma}^* C_{j\delta} \langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{AS} \} C_{i\beta} = \epsilon_i^{\text{HF}} C_{i\alpha}$:

The Hartree-Fock equations

$$\boxed{\sum_{\beta} h_{\alpha\beta}^{\text{HF}} C_{i\beta} = \epsilon_i^{\text{HF}} C_{i\alpha}} \quad h_{\alpha\beta}^{\text{HF}} = \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{j=1}^N \sum_{\gamma\delta} C_{j\gamma}^* C_{j\delta} \langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{AS}.$$

- A Hermitian eigenvalue problem, $\mathbf{h}^{\text{HF}} \mathbf{C} = \epsilon^{\text{HF}} \mathbf{C}$, with the one complication that **the matrix depends on its own eigenvectors**. In the eigenbasis of $\hat{h}_0$ the first term is $\epsilon_{\alpha} \delta_{\alpha\beta}$. No integrals are recomputed: $\langle \alpha | \hat{h}_0 | \beta \rangle$ and $\langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{AS}$ are tabulated once (n^4 storage), each iteration is a contraction with the current coefficients and a diagonalisation.

The density matrix and the self-consistent field (section 6.5)

The double sum over occupied orbitals enters only through the one-body **density matrix**

$$\rho_{\gamma\delta} = \sum_{i=1}^N \langle \gamma | i \rangle \langle i | \delta \rangle = \sum_{i=1}^N C_{i\gamma} C_{i\delta}^*, \quad \rho^2 = \rho, \quad \text{tr } \rho = N, \quad h_{\alpha\beta}^{\text{HF}} = \varepsilon_{\alpha} \delta_{\alpha\beta} + \sum_{\gamma\delta} \rho_{\gamma\delta} \langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{\text{AS}},$$

the projector onto the occupied subspace (the invariance of section 6.3, formalised).

The self-consistent field algorithm

1. Start from a guess, conventionally $C_{i\alpha}^{(0)} = \delta_{i\alpha}$ (the N lowest eigenstates of $\hat{h}_0$); random orthonormal vectors also work.
 2. Build ρ and h^{HF} ; diagonalise; take the N lowest eigenvectors as the new occupied orbitals.
 3. Repeat until the single-particle energies stop moving, $\frac{1}{m} \sum_p |\epsilon_p^{(n)} - \epsilon_p^{(n-1)}| \leq \lambda \sim 10^{-8}$.
- `hartreefock.py`, spinless fermions in a one-dimensional oscillator trap with a softened Coulomb repulsion, eight orbitals: $N = 4$ goes from 11.77084712 to $E^{\text{HF}} = 11.62305385$ in 17 iterations; the first diagonalisation gives 96% of the gain.
 - **Never use the energy as the convergence criterion.** At a stationary point the energy is quadratic in the orbital displacement: a drift of 10^{-5} in the ϵ_p is 10^{-10} in E . Everything that depends on the orbitals – f_{ai} , transition elements, the input of a coupled-cluster calculation – is only as good as the orbital drift (Figure 6.1: linear convergence, a factor four per iteration).

Brillouin's theorem, and FCI in the Hartree-Fock basis

At convergence the Fock matrix is diagonal in its own eigenbasis, so its occupied-unoccupied block vanishes:

$$f_{ai} = 0 \quad \Longleftrightarrow \quad \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0.$$

The Hartree-Fock solution is **the single-particle basis that decouples the reference from all single excitations**. Section 8 of the notebook, random Hamiltonian, $n = 8$, $N = 4$ (same E_0 in both bases, as it must be):

basis	E_{ref} or E^{HF}	CIS	CID	CISD	FCI
eigenstates of $\hat{h}_0$	6.497174	2.438662	5.442075	1.157492	0.747713
Hartree-Fock (48 iterations)	1.298016	1.298016	0.819738	0.774941	0.747713

- In the HF basis $\max |f_{ai}| = 9 \times 10^{-11}$, CIS returns E^{HF} exactly (Brillouin), the $0p0h-1p1h$ block of the FCI matrix is empty, and $\Delta E = -0.550$ is carried by the doubles term of the energy equation alone.
- Every truncation is better in the HF basis; the full CI is the same. **The truncations depend on the reference, the exact answer does not** – which is why Hartree-Fock comes first and every later method is built on it. And $E^{\text{HF}} = \sum_i \epsilon_i^{\text{HF}} - \frac{1}{2} \sum_{ij} \langle ij | \hat{v} | ij \rangle$: the sum of orbital energies double-counts the interaction (1.29801640 both ways in the notebook).

Pause to think: when the mean field does nothing

Pause to think #6

(about 3 min – discuss with your neighbour)

1. For the pairing model with the reference “levels 1 and 2 filled”, write down the Fock matrix f_{pq} (week 37, Example 5). What does the Hartree-Fock iteration do, and what is E^{HF} ?

Answer to pause #6

1. f is diagonal: $f_{ij} = \varepsilon_i - g/2$ for the holes (only the partner $\bar{i}$ contributes), $f_{aa} = \varepsilon_a$, $f_{ai} = 0$ because the interaction connects pairs to pairs only. The iteration converges at once: the reference *is* the Hartree-Fock determinant, $E^{\text{HF}} = E_{\text{ref}} = 2 - g$, and the whole correlation energy (-0.364 at $g = 1$) lies beyond the mean field. The pairing model is built so that Hartree-Fock does nothing and the pair correlations are the entire story.

Exercise session: week 38 (suggested set)

The set combines exercises from the chapter exercise sections of the lecture notes (where full answers are given) with new ones on this week's material. Chapter 5, exercises 1–3 (counting, vanishing blocks, the 6×6 pairing matrix) were assigned in week 37; if they were not done, they come first.

Exercise 1: bit patterns

- a) Five orbitals, bits $\alpha_0 \dots \alpha_4$. Evaluate $a_3^\dagger a_1 |01011\rangle$, $a_0^\dagger a_4 |01101\rangle$ and $a_4^\dagger a_3^\dagger a_1 a_0 |11010\rangle$ by anticommuting the operators into place, and again with the “occupied states strictly between” rule.
- b) Show that the phase of $a_j^\dagger a_i$ is the product of the phases of a_i and $a_j^\dagger$ taken separately, and that it equals $(-1)^l$ with l the number of occupied states strictly between i and j . Where does the argument use $i \neq j$?
- c) Count, for n orbitals and N particles, the number of index quadruples for which $a_p^\dagger a_q^\dagger a_s a_r |\Phi\rangle$ is non-zero, and compare with n^4 for $n = 40$, $N = 8$. Verify with `apply_hamiltonian` on random patterns.

Exercise 2: the FCI equation from the variational principle

- a) Starting from the Rayleigh quotient $E[C] = C^\dagger H C / C^\dagger C$, compute $\partial E / \partial C_K^*$ and show that the stationary points are the eigenvectors of H and the stationary values its eigenvalues. Repeat with a Lagrange multiplier and identify $\lambda = E$.
- b) For the two-level, one-pair system write down $\det(H - E1) = 0$, solve it, expand E_- to order g^2 and compare with second-order perturbation theory.
- c) Compute the characteristic polynomial of the 6×6 seniority-zero matrix at $g = 1$ (by hand for the trace and the determinant, with `numpy.poly` for the rest) and verify that its roots are the eigenvalues. Then do the same for the 70×70 matrix and explain what you see.
- d) (Interlacing.) Diagonalise the 6×6 matrix in the subspaces $\{|12\rangle\}$, $\{|12\rangle, |13\rangle\}$, $\{|12\rangle, |13\rangle, |14\rangle\}$, ... and verify that every eigenvalue decreases monotonically as the space grows and stays above the corresponding exact eigenvalue.

Exercise 3 (chapter 5, exercise 4): truncations and size consistency

- a) Restrict the pairing basis to excitation levels ≤ 1 and ≤ 2 , reproduce the CIS and CID columns, and explain why CIS returns exactly the reference energy.
- b) Build two non-interacting two-level pairing systems; show that full CI gives twice the energy of one while CID does not (tabulate the error for $g = 0.25, 0.5, 1$); identify the missing configuration, show that quadruples restore the exact result and that $\exp(\hat{T}_2)|\Phi_0\rangle$, with the parameters of CID, is size consistent.

Hint: 2a) treat C and C^* as independent.

What to do with the notebook

WeeklySlides/week38.ipynb contains, in this order (all self-contained numpy, mirroring fci.py and hartreefock.py):

1. Bit patterns: the creation-operator table, move against two single moves, $\hat{H}|\Phi\rangle$ by bits against the Fock-space matrices.
2. The pairing model by FCI: 70 determinants, the block table, 142 non-zero elements, 70×70 against 6×6 with the broken-pair amplitudes.
3. The variational principle: the Rayleigh quotient and its gradient, the Ritz bound, steepest descent, the characteristic polynomial for 6×6 and 2×2 , the degree-70 disaster.
4. The random $n = 8$, $N = 4$ Hamiltonian: the block table and the first column checked against E_{ref} , f_{ai} , $\langle ab|v|ij\rangle$, 0.
5. Truncations: the four-level table, six levels and six particles (924), size consistency with CID and CISDTQ, the one-parameter exponential ansatz.
6. The projected equations: every term of the energy, singles (eight) and doubles (thirteen) equations evaluated with the exact eigenvector, equations verified to 10^{-14} .
7. The iterative solution: first iterate $= E_{\text{ref}} - \frac{7}{24}g^2$, convergence to FCI.
8. Hartree-Fock: the SCF loop, $E_{\text{ref}} \rightarrow E^{\text{HF}} \rightarrow E_0$, Brillouin, the transformation to the HF basis and FCI/CIS/CID/CISD in it.

Play with it: change the seed and the interaction scale of the random Hamiltonian (what happens to the SCF loop when the interaction is strong?); run the iteration of section 7 at $g = 2$.

Summary of week 38

Full configuration interaction

- Determinants as integers; $\hat{H}|\Phi\rangle$ by bit operations; Ritz = optimise the coefficients; the multiplier is the energy; $HC = EC$, $\det(H - E1) = 0$
- Banded matrix from Condon-Slater: E_{ref} , f_{ai} , $\langle ab|v|ij\rangle$, 0; interlacing bounds
- Truncated CI: 99.8% $\rightarrow$ 77% of the correlation energy, not size consistent; $e^{\hat{T}}$ repairs it
- Energy, singles and doubles equations as diagrams; one disconnected diagram; iteration = perturbation theory

In code: `eigh` of a matrix built by bit operations, and a loop of `eigh` on a matrix that depends on its own eigenvectors – both in the notebook, both a few dozen lines.

Hartree-Fock

- Keep one determinant, optimise the orbitals: $E^{\text{HF}} \geq E_0$
- Constrained variation $\rightarrow h^{\text{HF}}C = \epsilon C$ with $h^{\text{HF}} = h_0 + \rho \cdot v$ depending on its own eigenvectors; iterate to convergence of the orbitals, never of the energy
- Brillouin: $f_{ai} = 0$; every truncation improves, FCI does not change

Next week, and a closing remark

Next week (week 39)

Hartree-Fock theory continued (chapter 6): the self-consistent field in practice with `hartreefock.py`; Hartree-Fock in second quantization – varying the determinant instead of the orbitals – and Thouless' theorem on what a nearby determinant is, with the Baker-Campbell-Hausdorff expansion it rests on; stability of the Hartree-Fock solution: stationary versus minimal, and the Lipkin model as an explicit instability; and, if time permits, the one system where the whole programme is analytic, the homogeneous electron gas. Reading: chapter 6, sections 6.6–6.11; Szabo and Ostlund chapter 3.

This week one equation did all the work. Written as $HC = EC$ it is the Ritz method and the exact answer for 70 or 924 determinants; written row by row it is the energy, singles and doubles equations, thirteen terms and thirteen diagrams; iterated, it is perturbation theory; truncated, it is CISD and the size-consistency defect that coupled cluster repairs; and read backwards – what basis makes its driving term vanish – it is Hartree-Fock theory. The variational principle produced both the expansion and the reference. The rest of the course is about combining the two.